



AlumniInsight

Technology Stack & Architecture Report

Integrated Alumni Analytics & Employability Prediction System
Faculty of Management Sciences — IIUI
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Frontend Technologies

TECHNOLOGY	VERSION / SOURCE	PURPOSE
HTML5	Vanilla	Page structure, semantic markup, single-page app layout
CSS3	Vanilla (styles.css)	Design system — dark mode, glassmorphism, gradients, micro-animations, responsive layout
JavaScript (ES6+)	Vanilla	Client-side logic — DOM manipulation, OOP classes, async/await, event handling
Chart.js	v4.4.1 (CDN)	Data visualization — bar, doughnut, radar, polar area, and line charts
Font Awesome	v6.5.1 (CDN)	Icon system — 100+ icons across the dashboard UI
Google Fonts	Inter, JetBrains Mono	Typography — primary UI font and monospace for code/data

⚙️ Backend Technologies

TECHNOLOGY	VERSION / SOURCE	PURPOSE
Python	3.x	Backend programming language
FastAPI	Latest	REST API framework — async endpoints, auto-docs, CORS middleware
Uvicorn	Latest	ASGI web server for serving FastAPI application
Pydantic	v2.x	Request/response data validation and serialization (BaseModel)
HTTPX	Latest	Async HTTP client for Supabase REST API and Groq API calls

🗄️ Database & Backend-as-a-Service

TECHNOLOGY	PURPOSE
Supabase (PostgreSQL)	Cloud-hosted relational database — alumni & companies tables (~4,000 records)
Supabase JS SDK v2	Frontend client — real-time queries, filtering, pagination, CRUD operations
Supabase REST API	Backend data access from FastAPI <code>m1-service.py</code> via HTTPX
LocalStorage / SessionStorage	Client-side authentication session persistence and user data caching



AI / Machine Learning

TECHNOLOGY	PURPOSE
Groq API (compound-beta)	Agentic AI assistant — intelligent chat, job market analysis, analytics widget generation
XGBoost	ML model for overall employability prediction (94.2% accuracy)
Random Forest	ML model for feature importance analysis (91.8% accuracy)
Neural Network (DL)	Deep Learning model for complex pattern recognition (93.5% accuracy)

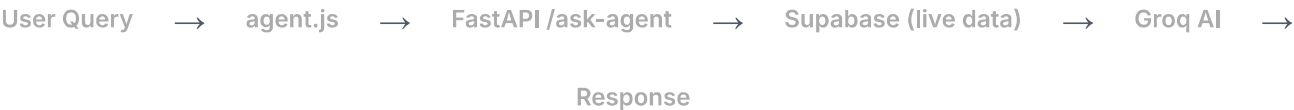


Architecture Overview

Design Patterns & Approach

Frontend Architecture:	Single-Page Application (SPA) — single index.html with JS-driven page routing
Authentication:	Client-side RBAC via AuthManager class (SHA-256 hashing, localStorage)
AI Integration:	Agentic RAG pattern — live data retrieval → context building → Groq reasoning
API Design:	RESTful endpoints with FastAPI auto-documentation (OpenAPI/Swagger)
Data Flow:	Supabase ↔ Frontend (JS SDK) + Supabase ↔ Backend (REST) ↔ Groq AI
Deployment:	Local dev server (frontend) + Uvicorn on port 8000 (backend)

AI Assistant Data Flow



🔧 API Endpoints (FastAPI)

METHOD	ENDPOINT	PURPOSE
GET	/	Service health — returns name, version, status
GET	/health	Connectivity check — Supabase + Groq status
POST	/ask-agent	AI chat — fetches live data, sends to Groq, returns answer + reasoning
GET	/stats	Direct access to aggregated alumni statistics
POST	/scrape-jobs	Job market intelligence — skill gap analysis via Groq
GET	/analytics-widgets	AI-powered dashboard insight cards

📁 Project File Structure

FILE	LINES	ROLE
index.html	1,961	Full SPA markup — auth pages, dashboard, 10+ page sections, AI panel
styles.css	~3,000+	Complete design system — dark/light mode, glassmorphism, animations
app.js	1,930	Dashboard logic — charts, navigation, auth UI, profile, admin panel
auth.js	601	AuthManager class — login, registration, RBAC, session management
supabase.js	272	AlumniDB class — Supabase CRUD, queries, filters, CSV export
agent.js	543	AgentManager class — AI assistant panel, Groq integration, reasoning
scraper.js	497	JobMarketScraper class — job intel, analytics widgets, chart rendering
ml-service.py	587	FastAPI backend — Supabase aggregation, Groq AI endpoints, CORS

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External Services & CDNs

SERVICE	URL / SOURCE	USAGE
Supabase Cloud	anxoxtavdydntwhavgdh.supabase.co	PostgreSQL database hosting + REST API
Groq Cloud API	api.groq.com	LLM inference — compound-beta model
jsDelivr CDN	cdn.jsdelivr.net	Chart.js + Supabase JS SDK delivery
cdnjs CDN	cdnjs.cloudflare.com	Font Awesome icon library
Google Fonts	fonts.googleapis.com	Inter & JetBrains Mono typefaces